

EU AI ACT

Governance, Risk, and Compliance for Computer Science and Society

Agenda

1. Why regulate AI?

2. EU AI Act architecture

3. Timeline and enforcement

4. Risk categories

5. High-risk systems

6. GPAI and generative AI

7. Compliance workflow

8. Case studies and discussion

Learning outcomes

- Explain the purpose and structure of the EU AI Act
- Distinguish prohibited, high-risk, transparency-risk, GPAI, and minimal-risk AI
- Apply the Act to realistic AI use cases in education, employment, finance, and public services
- Identify provider, deployer, importer, distributor, and user responsibilities
- Design a basic AI Act compliance plan for an AI system

**By the end: classify → justify →
mitigate → document**

Opening poll

ACTIVITY

Which AI use case worries you most from a societal perspective?

1. Facial recognition

2. AI hiring

3. AI tutors

4. Chatbots in public
services

PART 1

Why the EU AI Act exists

AI as infrastructure for the information society

From software to sociotechnical power

- AI is no longer only a technical artefact; it influences access, ranking, prediction, and decisions
- Modern AI systems shape information flows, institutional decisions, and everyday choices
- The EU AI Act treats AI as a sociotechnical system with risks for safety, health, and fundamental rights
- Regulation aims to support trustworthy innovation while limiting harmful uses

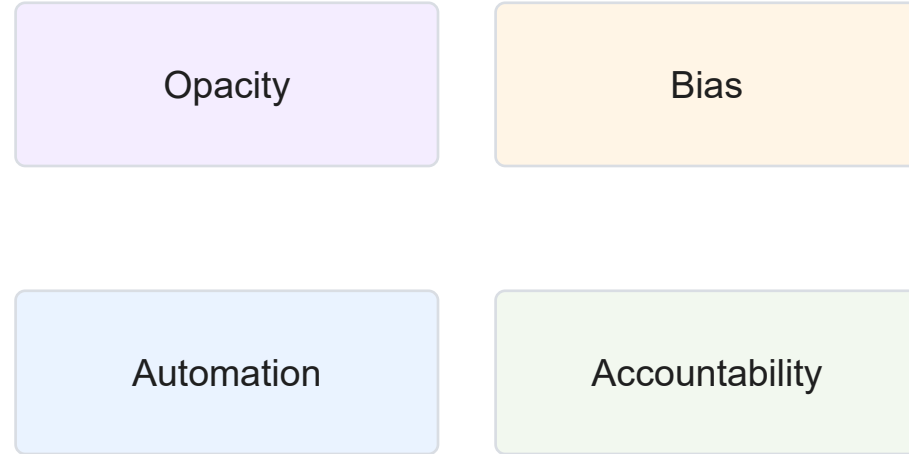
technical system

institutional use

social impact

What problem does the Act try to solve?

- Many AI risks are invisible until deployment: bias, opacity, automation bias, lack of contestability
- Market incentives do not always reward safety, documentation, fairness, or human oversight
- Citizens often cannot understand or challenge automated decisions
- A single EU framework can reduce fragmented national rules and support the internal market



A rights-based and market-based regulation

- Rights-based: protects fundamental rights, safety, dignity, privacy, equality, and democratic processes
- Market-based: creates harmonised rules for placing AI systems on the EU market
- Risk-based: obligations depend on the use case and risk level, not simply on the technology label
- Lifecycle-based: obligations continue from design and deployment to monitoring and incident reporting

Fundamental rights

Internal market

Risk management

Mini-discussion

ACTIVITY

Is AI regulation mainly about innovation, safety, or fundamental rights?

1. Think alone: 1 minute

2. Discuss in pairs: 4 minutes

3. Vote and justify

The AI Act in one sentence

- The AI Act is a horizontal EU regulation for AI systems, built on risk categories and lifecycle obligations
- It does not regulate every algorithm equally
- It regulates certain AI practices, systems, models, providers, deployers, and uses
- The main teaching challenge is not memorising articles, but learning how to classify use cases

Use case first, technology second

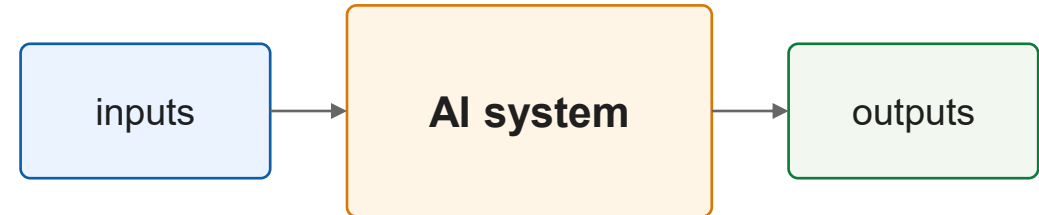
PART 2

Core definitions and scope

What counts as AI, who has obligations, and where the Act applies

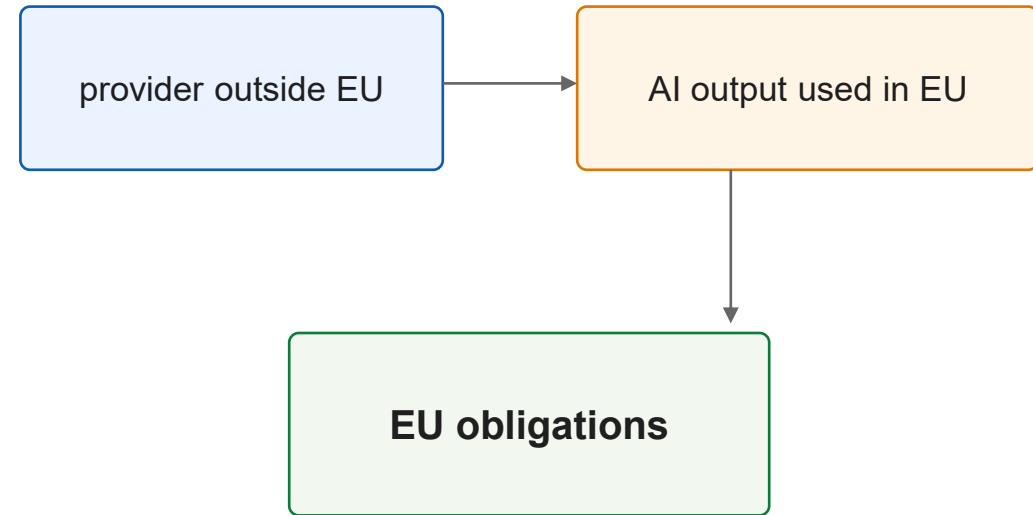
Key definition: AI system

- The Act defines an AI system broadly as a machine-based system designed to operate with varying levels of autonomy
- It may adapt after deployment and generate outputs such as predictions, content, recommendations, or decisions
- Outputs can influence physical or virtual environments
- The definition intentionally covers more than deep learning and generative AI



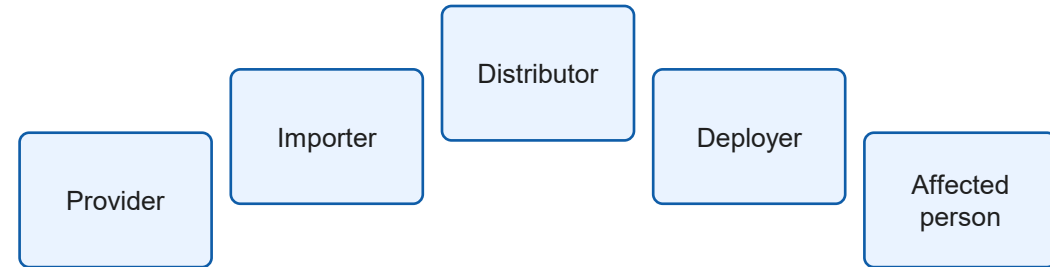
Scope: why non-EU actors may still be covered

- The Act applies to providers placing AI systems or GPAI models on the EU market
- It also applies when outputs are used in the EU, even if the provider is outside the EU
- Deployers in the EU are covered when they use AI systems in professional contexts
- This makes the Act relevant for global vendors, universities, startups, and public authorities



Actors in the AI Act

- **Provider:** develops or has an AI system developed and places it on the market or puts it into service
- **Deployer:** uses an AI system under its authority, except for purely personal non-professional activity
- **Importer and distributor:** bring AI systems into the EU supply chain
- **Product manufacturer:** integrates AI into regulated products such as medical devices or machinery



Technology-neutral, use-case-sensitive

- A chatbot can be low risk, transparency-risk, or high-risk depending on its purpose and context
- A classification model for credit scoring may be high-risk because it affects access to essential services
- An image generator may trigger transparency duties and GPAI obligations at model level
- A medical diagnosis system may be high-risk because it is safety-critical

**same technique
+ different context
= different legal result**

Classification warm-up

ACTIVITY

Classify each example: low, transparency, high, or prohibited.

1. AI spam filter

2. CV screening tool

3. Emotion recognition at
school

4. AI-generated image

PART 3

Timeline and implementation

What already applies and what is coming next

AI Act implementation timeline

- 1 Aug 2024: Act entered into force
- 2 Feb 2025: AI literacy and prohibited practices started applying
- 2 Aug 2025: GPAI rules and governance provisions started applying
- 2 Aug 2026: most remaining provisions apply, including many high-risk system duties
- 2 Aug 2027: Article 6(1) high-risk AI embedded in regulated products applies



What is already relevant in May 2026?

- AI literacy duties already apply
- Prohibited AI practices already apply
- GPAI provider obligations already apply for relevant models placed on the market from 2 Aug 2025
- Organisations should prepare now for high-risk obligations applying from 2 Aug 2026
- Legacy GPAI models placed on the market before 2 Aug 2025 have transitional considerations

NOW
May 2026

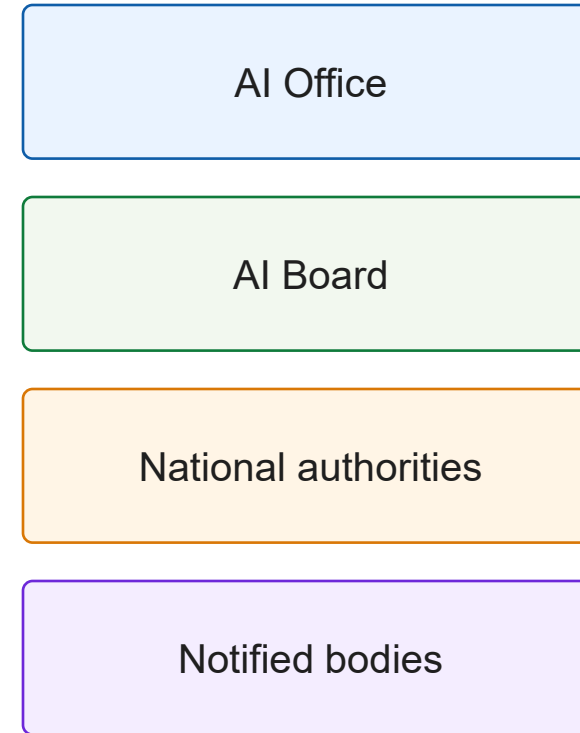
AI literacy: why it matters

- Article 4 requires providers and deployers to ensure a sufficient level of **AI literacy** among staff and other persons dealing with AI systems
- AI literacy is not only technical training; it includes capabilities, risks, context, and responsible use
- For universities and companies, this creates a practical need for structured training
- This lecture can be part of AI literacy capacity building



Governance actors

- European Commission and AI Office coordinate implementation, especially for GPAI
- AI Board supports consistent application across Member States
- National competent authorities supervise and enforce at Member State level
- Notified bodies play a role in conformity assessment for selected high-risk systems



Learning Control Question

Which duty already applies since 2 February 2025?

a) All high-risk conformity assessments

b) AI literacy and prohibited practices

c) Only GPAI systemic-risk duties

d) No duties until 2027

Correct answer: b

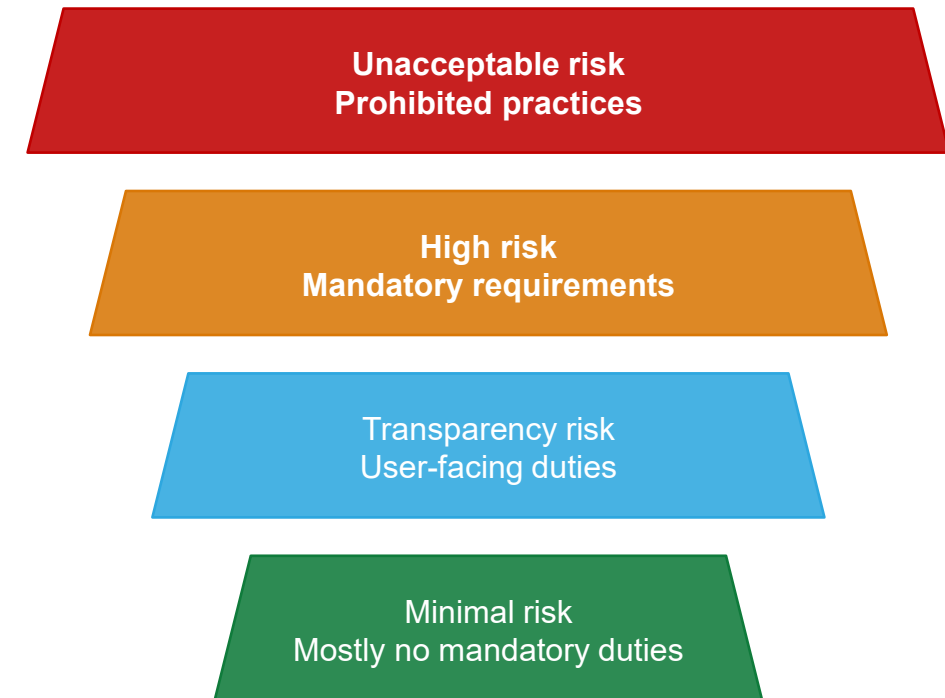
PART 4

Risk architecture

The backbone of the EU AI Act

Four-level risk logic

- Unacceptable risk: prohibited practices
- High risk: strict lifecycle requirements before and after deployment
- Specific transparency risk: disclose when people interact with certain AI or AI-generated content
- Minimal or low risk: generally no mandatory AI Act obligations, but voluntary codes may apply



Unacceptable risk: prohibited practices

- Manipulative or deceptive techniques that materially distort behaviour and cause or are likely to cause significant harm
- Exploitation of vulnerabilities due to age, disability, or social/economic situation
- Certain social scoring practices
- Certain biometric categorisation, emotion recognition, predictive policing, and scraping of facial images

BANNED
not merely “high risk”

Example: emotion recognition in education

- The Act prohibits AI systems used to infer emotions in workplace and educational institutions, except for medical or safety reasons
- This is directly relevant to EdTech and learning analytics
- Affective computing research must separate experimental research ethics from market deployment obligations
- Teachers and institutions need to understand when monitoring becomes inference about emotions

**Education + emotion
inference
→ generally prohibited**

High-risk category 1: product safety route

- AI is high-risk when it is a safety component of a product or is itself a product covered by listed EU harmonisation legislation
- Examples can include medical devices, machinery, toys, lifts, aviation, vehicles, and radio equipment
- Article 6(1) and corresponding obligations apply from 2 Aug 2027
- This route often links AI compliance with existing product safety regulation

**Regulated product
+ AI safety component
= high risk**

High-risk category 2: Annex III use cases

- Biometrics, if not prohibited
- Critical infrastructure
- Education and vocational training
- Employment, worker management, and access to self-employment
- Access to essential private and public services
- Law enforcement, migration/asylum/border control, justice and democratic processes



High-risk is not the same as “advanced AI”

- A simple model used for high-stakes decisions may be high-risk
- A sophisticated foundation model may be regulated through GPAI provisions even before deployment in a specific high-risk use case
- A very advanced recommender for entertainment may not be high-risk under the Act
- Classification depends on intended purpose, context, and impact

Risk = use + context + impact

Transparency obligations

- Users should know when they are interacting with certain AI systems, such as chatbots
- Certain AI-generated or manipulated content may require disclosure
- Emotion recognition and biometric categorisation systems trigger information duties where allowed
- Transparency duties are distinct from high-risk requirements

“Tell people when AI is involved”

Risk-ranking exercise

ACTIVITY

Rank four systems from least to most regulated.

1. Grammar checker

2. AI tutor

3. CV screening

4. Remote biometric ID

Learning Control Question

Which factor is most important for AI Act classification?

a) Programming language

b) Whether Python was used

c) Model parameter count only

d) Intended purpose and context

Correct answer: d

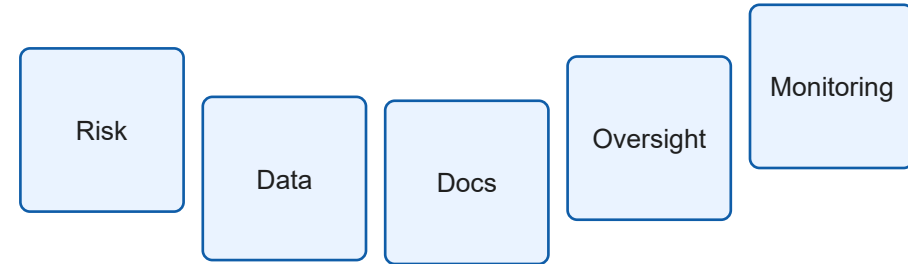
PART 5

High-risk AI systems

Requirements, roles, and operational meaning

Provider obligations for high-risk AI

- Risk management system throughout the lifecycle
- Data governance and data quality measures
- Technical documentation and record-keeping
- Transparency and instructions for use
- Human oversight, accuracy, robustness, and cybersecurity
- Post-market monitoring and serious incident reporting



Risk management system

- Identify reasonably foreseeable risks to health, safety, and fundamental rights
- Estimate and evaluate risks during design, testing, deployment, and post-market use
- Adopt mitigation measures and residual-risk controls
- Test against intended purpose and foreseeable misuse
- Document decisions so they can be audited

identify

evaluate

monitor

mitigate

Data governance

- Training, validation, and testing data must be relevant, sufficiently representative, and as free from errors as possible for intended purpose
- Bias management is a compliance issue, not just an ethical add-on
- Data provenance and preparation choices need to be documented
- Data governance is especially difficult when data reflect historical discrimination

data quality
→ **model behaviour**
→ **social outcome**

Technical documentation

- Documentation should show that the system complies with the Act
- It must describe intended purpose, design, development, testing, performance, risks, and controls
- Documentation supports conformity assessment and market surveillance
- For students: documentation is where engineering becomes accountable governance

Model card

Risk file

Instructions

Logs

Human oversight

- Human oversight is not a symbolic “human in the loop” checkbox
- Humans need authority, competence, information, and time to intervene
- Systems should be designed so humans can understand relevant outputs and limitations
- Automation bias is a key risk: humans may over-trust AI outputs

**meaningful oversight
≠ rubber stamp**

Accuracy, robustness, and cybersecurity

- High-risk systems must achieve appropriate levels of accuracy, robustness, and cybersecurity
- Performance should be measured against intended purpose and foreseeable conditions
- Robustness includes resilience to errors, faults, inconsistencies, and malicious attempts
- Cybersecurity matters because compromised AI systems can create physical, social, and legal harm

Accuracy

Robustness

Cybersecurity

Deployer obligations for high-risk AI

- Use the system according to instructions for use
- Assign human oversight to competent and trained persons
- Monitor operation and report serious incidents when relevant
- Keep logs where under their control
- Carry out fundamental rights impact assessments in specified public-service and high-stakes contexts

Deployers are not passive buyers

Fundamental Rights Impact Assessment

- A FRIA asks how a high-risk AI system may affect people and groups in context
- It should consider purpose, affected persons, risks, safeguards, human oversight, and complaints/appeal mechanisms
- This is especially relevant for public bodies and essential services
- It pushes organisations to analyse social consequences before deployment

**Who is affected?
What could go wrong?
How can people contest?**

High-risk workshop

ACTIVITY

Is an AI tool that predicts student dropout high-risk?

1. Identify intended purpose

2. Map affected persons

3. List benefits and harms

4. Decide controls

Learning Control Question

Which is NOT a core high-risk provider duty?

a) Risk management

b) Technical documentation

c) Data governance

d) Maximising ad engagement

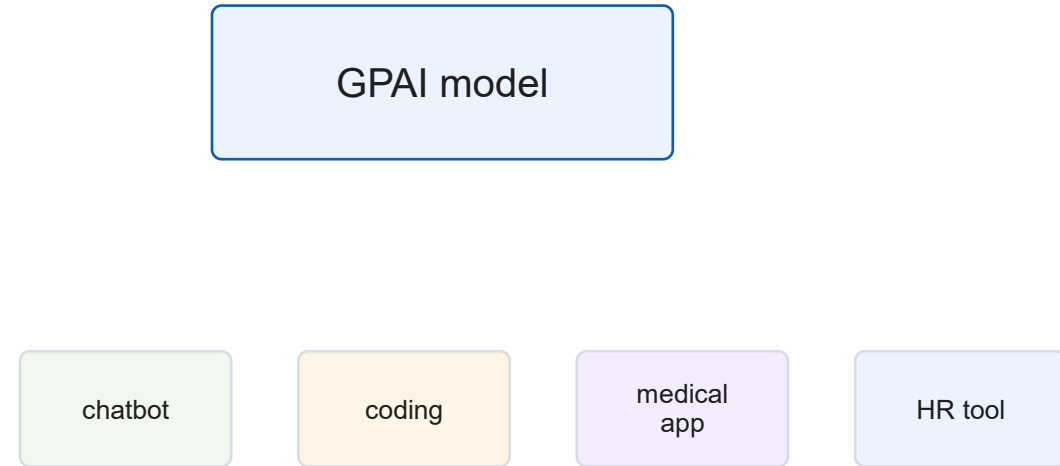
PART 6

GPAl, foundation models, and generative AI

What changed after ChatGPT

Why GPAI gets special treatment

- General-purpose AI models can be integrated into many downstream systems
- Risk is distributed across model providers, application developers, deployers, and users
- A single model can support both harmless and high-risk applications
- The Act therefore regulates GPAI models separately from specific AI systems



GPAI provider obligations

- Technical documentation for authorities and downstream providers
- Information to help downstream providers comply with their obligations
- Policy to respect EU copyright law
- Summary about content used for training, according to the template provided by the AI Office
- Additional duties apply to GPAI models with systemic risk

Transparency

Copyright

Downstream support

Systemic risk

GPAI Code of Practice

- The Commission published the General-Purpose AI Code of Practice on 10 July 2025
- It is a voluntary tool to help providers demonstrate compliance
- It has three main chapters: Transparency, Copyright, and Safety & Security
- Safety & Security is aimed at the most advanced models with systemic risk

Transparency

Copyright

Safety & Security

Generative AI and disclosure

- People should know when they interact with certain AI systems, especially chatbots
- Synthetic or manipulated content may require disclosure in relevant cases
- Deepfakes create special trust and democratic risks
- Disclosure alone is not enough; it must be paired with media literacy and platform governance

content looks real
→ **trust problem**
→ **disclosure duty**

Open-source and research questions

- The AI Act contains important provisions and exemptions for research, development, testing, and some open-source models
- But open-source status does not automatically remove all obligations
- Downstream deployment can still create high-risk or transparency-risk duties
- Students should ask: who releases the model, who modifies it, who deploys it, and for what purpose?

Open source is a governance question, not a magic exemption

AI Act vs generative AI hype

- The Act is not only about ChatGPT-style systems
- It also covers predictive, classificatory, biometric, recommendation, and decision-support AI
- Generative AI makes transparency, copyright, and systemic-risk questions more visible
- But the high-risk regime often concerns ordinary decision systems used in sensitive domains

**Do not reduce AI
governance to chatbots**

GPAI role-play

ACTIVITY

A university wants to deploy a GPAI chatbot for student advising. Who has which responsibilities?

1. Model provider

2. App developer

3. University deployer

4. Students affected

Learning Control Question

Which three chapters structure the GPAI Code of Practice?

a) Speed, cost, usability

b) Transparency, Copyright, Safety & Security

c) Privacy, marketing, sales

d) Hardware, software, networks

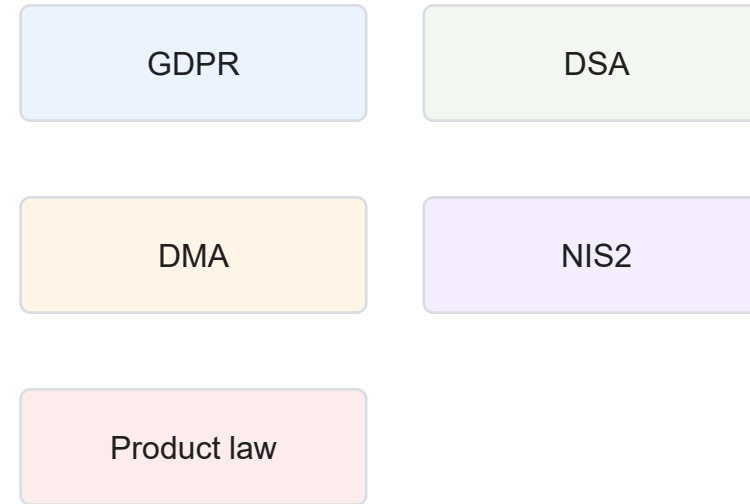
PART 7

AI Act and other EU digital rules

Where the Act fits in the wider governance landscape

AI Act is not alone

- GDPR (General Data Protection Regulation): personal data, lawful basis, data subject rights, automated decision-making
- DSA (Digital Services Act): online platforms, systemic risks, recommender transparency, illegal content
- DMA (Digital Markets Act): gatekeeper platforms and contestability
- Cybersecurity rules such as NIS2 (Network and Information Security Directive 2) can matter when AI is part of critical services
- Product safety and sector-specific law can still apply



AI Act and GDPR

- The AI Act does not replace GDPR
- If AI uses personal data, GDPR requirements still apply
- Bias mitigation can create tension with data minimisation and sensitive data rules
- Automated decisions affecting people may require GDPR analysis in parallel
- Compliance teams need joint legal, technical, and organisational workflows

**AI compliance
+ data protection
= joint governance**

AI Act and education

- AI tutoring, grading, admissions, proctoring, and dropout prediction can affect learning opportunities
- Annex III includes education and vocational training use cases
- Emotion recognition in educational institutions is generally prohibited except medical/safety reasons
- Universities need policies for procurement, use, explainability, appeal, and AI literacy

**EdTech is not only innovation.
It is also rights, evidence, and
accountability.**

AI Act and employment

- Recruitment, selection, candidate ranking, promotion, termination, and worker monitoring can be high-risk
- The core societal risk is access to livelihood and unequal treatment
- Documentation must address data quality, bias, oversight, and contestability
- HR deployers need trained staff and clear accountability

**Hiring AI = high-stakes allocation
of opportunity**

AI Act and public services

- AI for eligibility, prioritisation, or access to essential services can be high-risk
- Public bodies may need fundamental rights impact assessments
- Citizens need transparency, oversight, and contestability
- This links AI governance with democratic legitimacy and administrative fairness

Public-sector AI must be explainable to society, not only to engineers

Comparative law map

ACTIVITY

For one AI system, identify which rules might apply besides the AI Act.

1. GDPR?

2. DSA/DMA?

3. NIS2?

4. Sector law?

PART 8

Compliance as an engineering workflow

How to turn legal categories into practical steps

A 5-step AI Act workflow

- 1. Inventory AI systems and GPAI dependencies
 - 2. Define intended purpose and actors
 - 3. Classify risk category
 - 4. Identify obligations and control
 - 5. Perform risk, data, and rights assessments
-
- 6. Implement controls, documentation, and oversight
 - 7. Monitor, log, update, and report incidents

Step 1: AI inventory

- List all AI systems, models, vendors, APIs, datasets, and intended uses
- Include shadow AI: tools used by staff without formal procurement
- Record owner, purpose, users, affected persons, data types, and vendor dependencies
- Prioritise systems affecting people in sensitive domains

**No inventory → no
governance**

Step 2: intended purpose

- Intended purpose is central to classification and documentation
- Describe what the system is meant to do, for whom, in which context, and with which limits
- Avoid vague statements such as “supports decision-making” without domain details
- Foreseeable misuse should also be considered

**What decision?
For whom?
In what context?**

Step 3: risk classification matrix

- Check prohibited practices first
- Then check high-risk routes: Article 6(1) product route and Annex III use cases
- Then check GPAI and transparency duties
- Finally decide whether minimal-risk/voluntary obligations remain
- Document the reasoning even when classification is “not high-risk”

Prohibited?

High-risk?

GPAI?

Transparency?

Minimal?

Step 4: control design

- Legal requirements must become design and process controls
- Examples: human override, confidence thresholds, audit logs, bias testing, user notices, model cards
- Controls should be tested, maintained, and updated
- A control that exists only in a policy document is weak governance

**Requirement → Control →
Evidence**

Step 5: evidence package

- Technical documentation
- Data governance records
- Risk management file
- Instructions for use and user information
- Human oversight procedures
- Testing and validation reports
- Post-market monitoring plan and incident response procedure

**Evidence is what makes
compliance auditable**

Compliance sprint

ACTIVITY

Build a one-page compliance plan for an AI admission-screening tool.

1. Classify risk

2. Identify actors

3. List obligations

4. Define evidence

PART 9

Case studies for classroom use

Applying the AI Act to realistic systems

Case 1: AI tutoring system

- Scenario: a university deploys an AI tutor that gives hints, feedback, and study recommendations
- Questions: Is it only learning support, or does it determine access/admission/evaluation?
- Risks: hallucination, unequal support, privacy, over-reliance, hidden profiling
- Possible duties: transparency, AI literacy, data protection, and possibly high-risk if used for assessment or access

Classification depends on educational consequence

Case 2: AI grading assistant

- Scenario: an AI system scores essays and recommends grades to teachers
- Education and evaluation use cases can fall into Annex III high-risk categories
- Human oversight must be meaningful, not merely formal approval
- Students need transparency and routes to challenge decisions

**High stakes for learning
progression**

Case 3: HR CV screening

- Scenario: a company ranks applicants using machine learning
- Employment recruitment and selection is an Annex III high-risk domain
- Provider obligations: data governance, risk management, documentation, oversight design
- Deployer obligations: use according to instructions, trained oversight, monitoring, possible FRIA

Opportunity allocation
→ **high-risk domain**

Case 4: Customer-support chatbot

- Scenario: a bank uses a chatbot to answer customer questions
- Usually transparency duties are central: users should know they interact with AI
- If chatbot affects access to essential services or credit decisions, risk level may increase
- Hallucination controls and escalation to humans are important safeguards

Chatbot risk rises when it affects rights or access

Case 5: Emotion recognition in online proctoring

- Scenario: a proctoring tool claims to detect stress, deception, or suspicious emotions
- Emotion inference in educational institutions is generally prohibited except medical/safety reasons
- Even separate from the AI Act, privacy and fairness concerns are substantial
- Alternative: use proportionate, transparent, non-emotion-based academic integrity measures

Likely prohibited if used for emotion inference in education

Case 6: Medical triage AI

- Scenario: AI prioritises patients based on symptoms and history
- May be high-risk through product safety route or sensitive essential-service use
- Requires robust validation, human oversight, cybersecurity, and post-market monitoring
- Sector-specific medical-device rules may apply in parallel

**Safety-critical + health
context**

Case-study presentations

ACTIVITY

Each group presents one classification decision in 3 minutes.

1. Facts

2. Classification

3. Obligations

4. Open questions

PART 10

Critical perspectives and debate

Critique 1: Innovation burden

- Some organisations argue that compliance may slow product development and increase costs
- Small providers may struggle with documentation and conformity assessment resources
- Supporters argue that trustworthy AI can become a competitive advantage
- Teaching point: regulation creates both constraints and design requirements

**Burden or trust
infrastructure?**

Critique 2: Legal uncertainty

- Classification can be difficult for hybrid and rapidly changing systems
- Many details depend on standards, guidelines, national authorities, and future enforcement practice
- The same system can change category if its intended purpose changes
- Students should learn to reason under uncertainty and document assumptions

**Uncertainty ≠ no action.
It means document your
reasoning.**

Critique 3: Global competition

- EU regulation interacts with US, UK, China, and international AI governance approaches
- Global providers may adapt products and documentation for multiple jurisdictions
- Europe's strategic question: can trustworthy AI become a strength rather than a bottleneck?
- This creates interdisciplinary opportunities for computer scientists

Computer scientists need legal literacy; lawyers need technical literacy

Structured debate

ACTIVITY

Motion: “The EU AI Act will improve AI innovation in Europe.”

1. Pro side: 5 min prep

2. Con side: 5 min prep

3. Debate: 10 min

4. Reflection vote

What computer science students should take away

- AI governance is now part of responsible software engineering
- Risk classification begins with intended purpose and societal context
- Documentation, testing, monitoring, and oversight are technical as well as legal tasks
- Current AI topics—GPAI, generative AI, deepfakes, AI tutors—must be analysed through risk and rights
- The EU AI Act is not the end of ethics; it is a minimum governance baseline

Build systems that can be explained, audited, and contested

Learning Control Question

Which statement is most accurate?

- a) The AI Act bans all AI in education
- b) The AI Act only regulates generative AI
- c) The AI Act is risk-based and use-case-sensitive
- d) The AI Act replaces GDPR

PART 11

Wrap-up and assessment

Review, questions, and transfer task

Review: the whole lecture in five ideas

- The EU AI Act is horizontal, risk-based, and rights-oriented
- Classification depends on intended purpose, context, actors, and impact
- Prohibited practices and AI literacy already apply; GPAI rules also apply
- High-risk AI needs lifecycle governance: risk, data, documentation, oversight, monitoring
- Computer scientists need to translate legal duties into system design and evidence

**classify → comply →
document → monitor**

Transfer task

- Choose one AI system from education, employment, healthcare, finance, public administration, or social media
- Describe the intended purpose, users, affected persons, and data
- Classify it under the AI Act risk logic
- Identify provider and deployer responsibilities
- Propose at least five technical or organisational safeguards
- Prepare a 5-slide presentation with sources

Deliverable: 5-slide AI Act classification report

Suggested 5-slide student report structure

- Slide 1: AI system and intended purpose
- Slide 2: Actors, data, users, and affected persons
- Slide 3: Risk classification and justification
- Slide 4: Obligations and safeguards
- Slide 5: Open questions, limitations, and sources

Slide 1

Slide 2

Slide 3

Slide 4

Slide 5

Learning Control Question

What should students do when the classification is uncertain?

a) Ignore the Act

b) Document assumptions and seek legal/technical review

c) Always classify as minimal risk

d) Only ask the model vendor

Final reflection

- What is one AI system you now see differently after learning about the AI Act?
- Which obligation seems most difficult to implement in practice?
- Where should responsibility sit: model provider, deployer, user, or regulator?
- What should computer science education change because of the AI Act?

Think: 2 min
Discuss: 5 min
Share: 8 min

Selected sources

- EU AI Act Service Desk, Timeline for implementation of the EU AI Act, accessed May 2026.
- European Commission, AI Act | Shaping Europe's digital future, updated 2026.
- AI Act Service Desk, Article 113: Entry into force and application, Regulation (EU) 2024/1689.
- European Commission, Guidelines on prohibited AI practices under the AI Act, 4 Feb 2025.
- European Commission, General-Purpose AI Code of Practice, published 10 Jul 2025.

Selected AI Act articles for teaching

- Article 3: definitions, including AI system
- Article 4: AI literacy
- Article 5: prohibited AI practices
- Article 6 and Annex III: high-risk classification
- Articles 9–17: requirements for high-risk AI systems
- Articles 26–27: deployer duties and fundamental rights impact assessment
- Article 50: transparency obligations
- Articles 51–56: GPAI models
- Article 113: entry into force and application

THANK YOU

Questions and discussion